

# **Lab 1 | NIST CSF & Incident Response**

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**ISEC3078 – Incident Handling & Response**

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## Objective

In this lab, you will simulate a small security incident on a Linux virtual machine and apply the **NIST Cybersecurity Framework (CSF)** by **detecting, responding to,** and **recovering from** suspicious login activity.

You will focus on the **Detect, Respond, and Recover** functions.

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## Environment

- Ubuntu Linux VM 24.04.3 (VirtualBox or VMware)
  - Terminal access
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## Scenario

You are monitoring a Linux system when suspicious login activity occurs.

Your task is to **create, detect,** and **respond to** this activity like an Incident Response analyst.

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## Part 1 | Generate Failed Login Attempts (Setup)

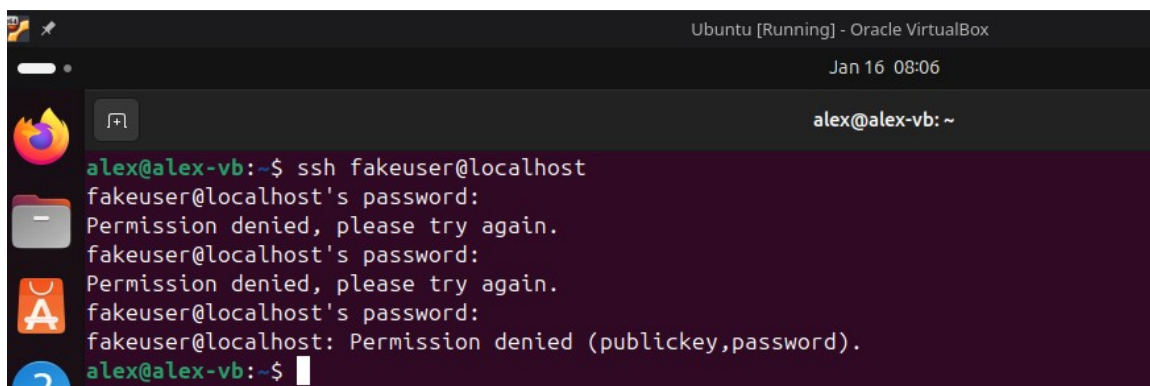
You will intentionally generate failed login attempts on **your own VM** for learning purposes.

### Steps

1. Open **Terminal**.
2. Install **ssh** if it is not already
3. Run the following command:

***ssh fakeuser@localhost***

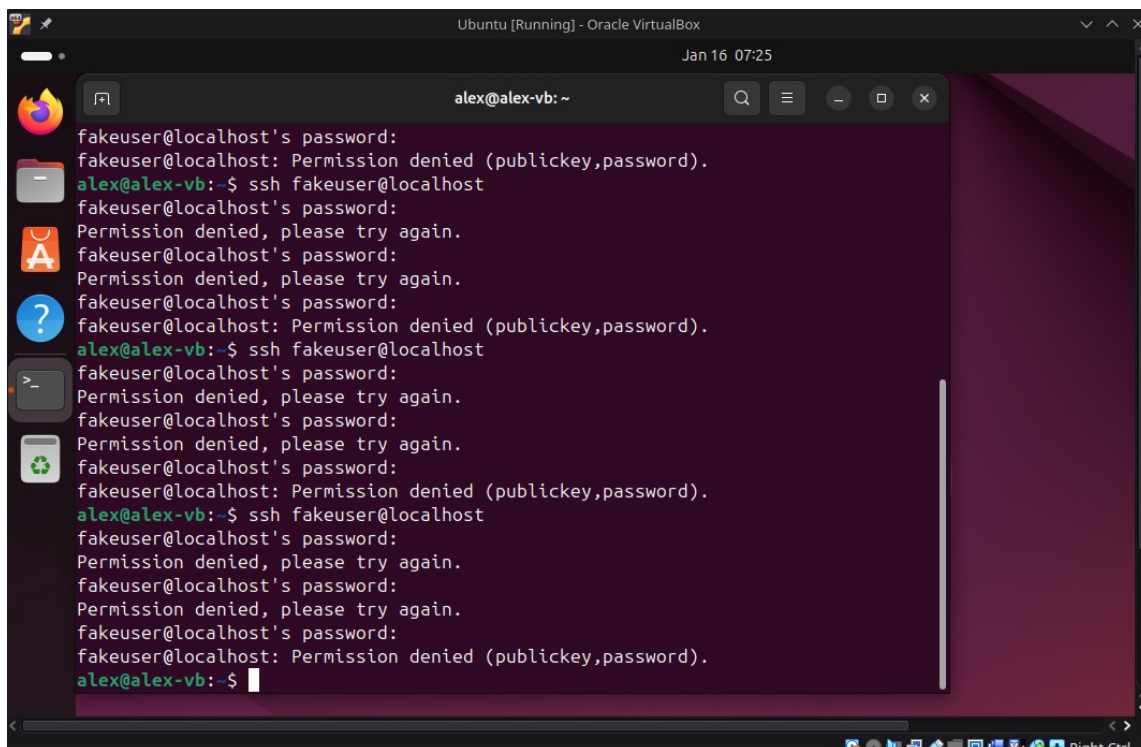
4.



```
Ubuntu [Running] - Oracle VirtualBox
Jan 16 08:06
alex@alex-vb: ~
alex@alex-vb:~$ ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
alex@alex-vb:~$
```

When prompted, enter **any incorrect password**.

5. Repeat this **3–5 times**.

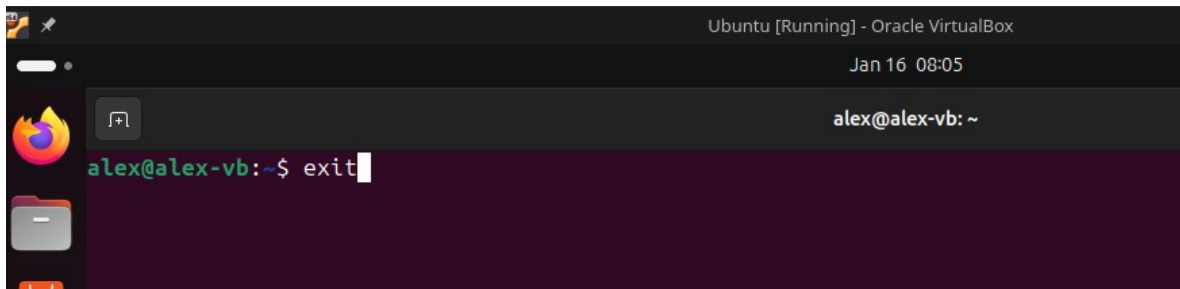


```
Ubuntu [Running] - Oracle VirtualBox
Jan 16 07:25
alex@alex-vb: ~
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
alex@alex-vb:~$ ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
alex@alex-vb:~$ ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
alex@alex-vb:~$ ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
alex@alex-vb:~$
```

6.Exit the SSH attempt if needed:

***exit***

This



simulates suspicious login behavior and creates log entries.

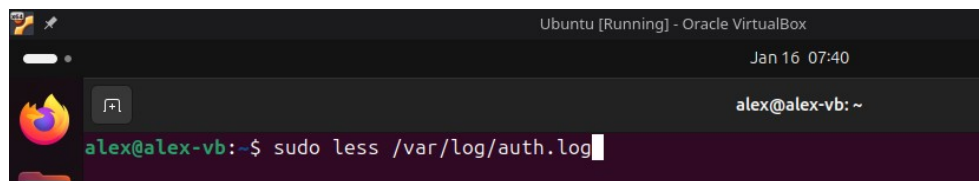
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## Part 2 | CSF Detect – Find the Logs

### Steps

1. In Terminal, view authentication logs:

***sudo less /var/log/auth.log***



2. Look for entries such as:

- **Failed password**
- **invalid user**
- **authentication failure**

```
session opened for user root(uid=0) by root(uid=0)
2026-01-16T08:30:44.697761-04:00 alex-vb CRON[3845]: pam_unix(cron:session): session closed for user root
2026-01-16T08:30:45.841305-04:00 alex-vb sshd[3838]: Failed password for invalid user fakeuser from 127.0.0.1 port 52050 ssh2
2026-01-16T08:30:48.359834-04:00 alex-vb sshd[3838]: pam_unix(sshd:auth): check pass; user unknown
2026-01-16T08:30:50.919595-04:00 alex-vb sshd[3838]: Failed password for invalid user fakeuser from 127.0.0.1 port 52050 ssh2
2026-01-16T08:30:51.527432-04:00 alex-vb sshd[3838]: Connection closed by invalid user fakeuser 127.0.0.1 port 52050 [preauth]
2026-01-16T08:30:51.529074-04:00 alex-vb sshd[3838]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
2026-01-16T08:31:07.301213-04:00 alex-vb sudo:      alex : TTY=pts/0 ; PWD=/home/alex ; USER=root ; COMMAND=/usr/bin/less /var/logs/auth.log
2026-01-16T08:31:07.301786-04:00 alex-vb sudo: pam_unix(sudo:session): session opened for user root(uid=0) by alex(uid=1000)
2026-01-16T08:31:07.307367-04:00 alex-vb sudo: pam_unix(sudo:session): session closed for user root
2026-01-16T08:31:14.877939-04:00 alex-vb sudo:      alex : TTY=pts/0 ; PWD=/home/alex ; USER=root ; COMMAND=/usr/bin/less /var/log/auth.log
2026-01-16T08:31:14.879361-04:00 alex-vb sudo: pam_unix(sudo:session): session opened for user root(uid=0) by alex(uid=1000)
~
~
(END)
```

3. Identify at least **one failed login attempt** you created.

**2026-01-16T08:30:50.915595-04:00 alex-vb sshd[3838]: Failed password for invalid user fakeuser from 127.0.0.1 port55948 ssh2**

## Answer

- **Why is this activity suspicious?**

Multiple failed login attempts in quick succession indicates that somebody is trying to “brute-force” access by guessing the correct credentials.

- **Which CSF function does this represent?**

This represents the **Detect** function, since log monitoring is being used to identify potential security events and threats.

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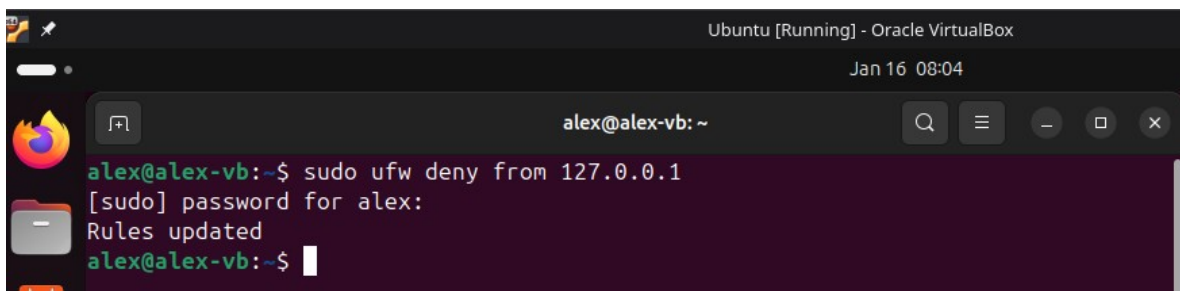
## Part 3 | CSF Respond – Contain the Activity

### Steps

1. Identify the IP address shown in the failed login entries.
2. Block the IP using the firewall:

***sudo ufw deny from <IP\_ADDRESS>***

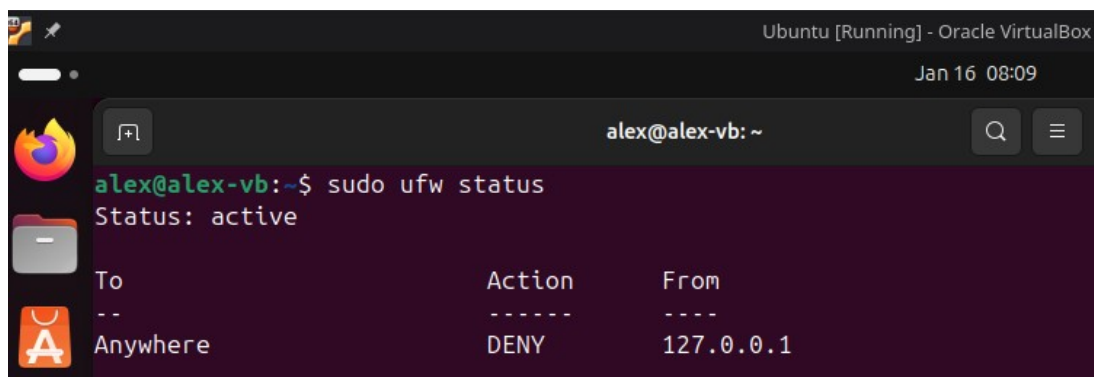
3.



```
alex@alex-vb:~$ sudo ufw deny from 127.0.0.1
[sudo] password for alex:
Rules updated
alex@alex-vb:~$
```

Verify the rule:

***sudo ufw status***



```
alex@alex-vb:~$ sudo ufw status
Status: active

To Action From
--
Anywhere DENY 127.0.0.1
```

### Answer

- **Why is this a response action?**

Taking note of the failed login attempts from 127.0.0.1, measures were taken to prevent further attempts by adding a firewall rule denying traffic to the IP address. This is a response action because it is a direct reply to a detected anomaly.

- **How does it limit further risk?**

With an active firewall rule blocking traffic from the source of the suspicious action, that traffic will not be able to make further attempts.

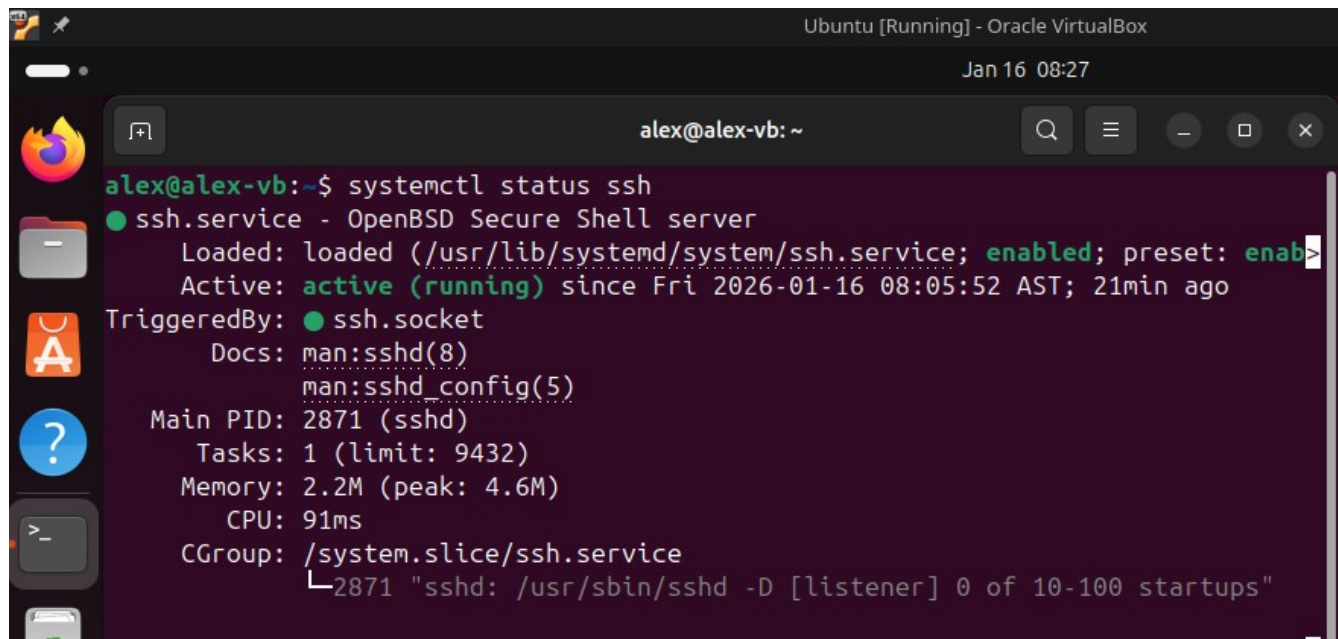
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## Part 4 | CSF Recover – System Readiness

### Steps

1. Check the SSH service status:

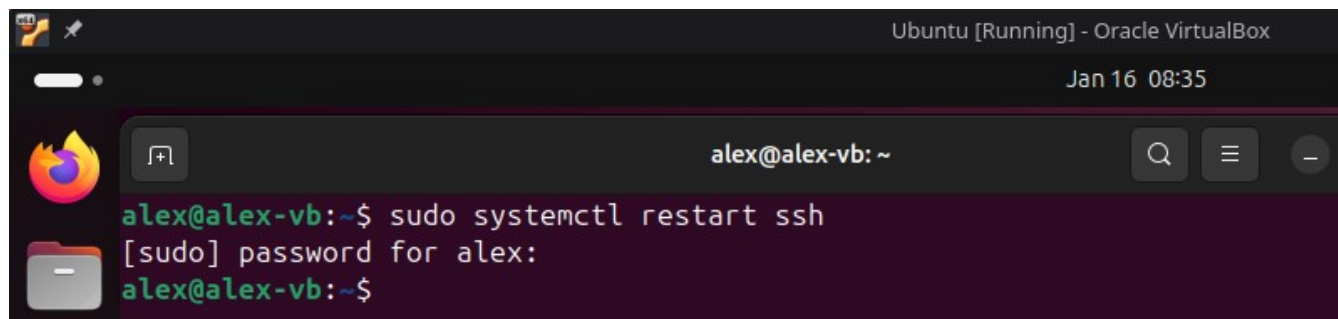
***systemctl status ssh***

A terminal window titled 'alex@alex-vb: ~' showing the command 'systemctl status ssh' and its output. The output indicates that the ssh.service is loaded, enabled, and active (running) since Fri 2026-01-16 08:05:52 AST, 21 minutes ago. It also shows the service's configuration, including its main PID (2871), tasks, memory usage, CPU usage, and CGroup.

```
alex@alex-vb:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>
   Active: active (running) since Fri 2026-01-16 08:05:52 AST; 21min ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 2871 (sshd)
      Tasks: 1 (limit: 9432)
     Memory: 2.2M (peak: 4.6M)
        CPU: 91ms
      CGroup: /system.slice/ssh.service
             └─2871 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

2. Restart the service if needed:

***sudo systemctl restart ssh***

A terminal window titled 'alex@alex-vb: ~' showing the command 'sudo systemctl restart ssh'. The terminal prompts for the password for 'alex' and then shows the command being executed successfully.

```
alex@alex-vb:~$ sudo systemctl restart ssh
[sudo] password for alex:
alex@alex-vb:~$
```

### Answer

- **How does this step support recovery?**

Recovery is supported through ensuring that SSH is working properly (and restarting it if it isn't). This is a deliberate action taken to restore functionality to a secure and familiar state after a potential threat. The verification of its status confirmed that it was operational, and that it was not adversely affected by the new firewall rule. In addition, any potential residual connection to the attacker IP was eliminated by restarting the service.

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## Part 5 | Incident Summary

Write a short incident summary (5–7 sentences) that includes:

- What activity was detected
- How it was responded to
- How the system was recovered
- One lesson learned to prevent future incidents

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A **large amount of failed login attempts** were detected via SSH from the IP address **127.0.0.1**. Multiple log entries were detected indicating “*Failed password*” and “*Authentication failure*”. This is an indicator of a potential brute-force attack on an account with the username “**fakename**”. Using the host firewall, the source IP address was identified and immediately blocked. By blocking the IP address, stopping further suspicious login attempts. To ensure recovery, SSH status was verified and restarted to ensure that it entered a functional operational state and clear of any possible remaining connection to the attacker IP. For future detection and prevention, regular review of authentication logs, establishing alerts for repeat failed attempts, and the use of an automatic blocking tool such as **fail2ban** for Ubuntu is recommended to detect and block traffic after numerous failed logins.

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